Concentration Inequalities: 20 Worked Problems

Markov · Chebyshev · Jensen · WLLN · CLT

These five tools form a ladder of increasingly strong (and increasingly assumption-hungry) guarantees: Markov needs only a mean, Chebyshev adds variance, Jensen adds convexity structure, and the limit theorems (WLLN, CLT) trade finite-sample rigor for the power of large n. Every problem is color-coded by tool and carries an analogy — and several problems deliberately reuse the same scenario across tools so you can watch the bounds get dramatically tighter as you add more assumptions.

A. Markov's Inequality

Problem A1 — The basic bound

A trading desk's daily loss $X \geq 0$ (in \$k) has mean $\mathbb{E}[X] = \$500$k. Bound $P(X \geq 2000)$ using only this mean.

Solution.

$$P(X \geq 2000) \leq \frac{\mathbb{E}[X]}{2000} = \frac{500}{2000} = \boxed{0.25}.$$

★ *Intuition:* Markov's inequality is the "worst-case budget" bound: if the average loss is \$500k, losses can't *routinely* be far above that without dragging the average up — but it makes zero use of how losses are actually spread out, which is why it will turn out to be extremely conservative (see A4).

Problem A2 — Where Chebyshev actually comes from

Let $Y = (X - \mu)^2$ where X has variance $\sigma^2 = 100$. Apply Markov's inequality to the nonnegative variable Y to bound $P((X - \mu)^2 \geq 400)$.

Solution. $Y \geq 0$ and $\mathbb{E}[Y] = \sigma^2 = 100$:

$$P(Y \geq 400) \leq \frac{\mathbb{E}[Y]}{400} = \frac{100}{400} = \boxed{0.25}.$$

Equivalently, $P(|X - \mu| \geq 20) \leq 0.25$ — with $k = 20/\sigma = 2$, this is exactly $1/k^2 = 1/4$.

★ *Intuition:* This *is* Chebyshev's inequality — it's not a separate tool, it's Markov applied to the squared deviation. Every "stronger" inequality in this chapter is secretly Markov in a clever disguise; recognizing the disguise is worth more than memorizing each one as an independent formula.

Problem A3 — Markov's bound can be exactly tight

Construct a nonnegative random variable with mean 10 for which the Markov bound $P(X \geq 20) \leq 0.5$ holds with *equality*.

Solution. Let $X = 20$ with probability 0.5 and $X = 0$ with probability 0.5. Then $\mathbb{E}[X] = 10$, and

$$P(X \geq 20) = 0.5 = \frac{\mathbb{E}[X]}{20}. \quad \checkmark \text{ Exactly tight.}$$

★ *Intuition:* This proves Markov's inequality cannot be improved *in general* — given only a mean, there genuinely exists a distribution (a two-point spike) that makes the bound exact. Any tighter general-purpose bound *must* use more information than the mean alone (variance, shape, support), which is exactly the next three sections' entire reason for existing.

Problem A4 — How loose is "loose"? A concrete comparison

$X \sim \text{Exp}(\lambda{=}1)$, so $\mathbb{E}[X] = 1$. Compare the Markov bound on $P(X \geq 5)$ to the *exact* value.

Solution.

$$\text{Markov: } P(X \geq 5) \leq \frac{1}{1} \cdot \frac{1}{5} = \boxed{0.2}.$$

$$\text{Exact: } P(X \geq 5) = e^{-5} \approx \boxed{0.0067}.$$

$$\text{Markov overestimates by a factor of } \approx \boxed{30\times}.$$

★ *Intuition:* This is the honest price of Markov's generality: it's valid for *every* nonnegative distribution with this mean, so it has to be conservative enough to cover worst-case distributions like A3's two-point spike — even when your actual distribution (Exponential, here) has a much thinner tail. Use Markov when you know nothing else; upgrade immediately once you know more.

B. Chebyshev's Inequality

Problem B1 — The basic bound

A strategy's daily return has mean 0.1% and sd 2%. Bound $P(|\text{return} - 0.1\%| \geq 6\%)$.

Solution. $k = 6/2 = 3$:

$$P(|X - \mu| \geq 3\sigma) \leq \frac{1}{k^2} = \frac{1}{9} \approx \boxed{0.111}.$$

★ *Intuition:* Chebyshev converts "how many standard deviations away" directly into a universal probability cap, valid for *any* distribution with finite variance — no bell-curve assumption required. That universality is both its strength and the reason it's looser than CLT-based bounds (Section E) whenever normality actually holds.

Problem B2 — A distribution-free confidence interval, and its price

Without assuming any particular distribution, find k such that $[\mu - k\sigma, \mu + k\sigma]$ contains X with probability at least 95%. Compare k to the Normal distribution's corresponding value.

Solution. Need $1/k^2 \leq 0.05 \implies k \geq \sqrt{20} \approx \boxed{4.47}$. Compare to the Normal's $z = 1.96$ for a 95% interval:

$$\frac{4.47}{1.96} \approx \boxed{2.28\times} \text{ wider.}$$

★ *Intuition:* This is the concrete cost of distribution-freedom: giving up the normality assumption more than doubles the width of your guaranteed-coverage interval. Every time you see a "distribution-free" or "worst-case" guarantee in quant finance, expect it to be this much more conservative than the Gaussian-based version everyone actually uses in practice.

Problem B3 — Chebyshev on a sample mean (the WLLN engine)

$n = 100$ iid trades, each with variance $\sigma^2 = 400$. Bound $P(|\bar{X} - \mu| \geq 5)$.

Solution. $\text{Var}(\bar{X}) = \sigma^2/n = 400/100 = 4$, so $\text{sd}(\bar{X}) = 2$, and $k = 5/2 = 2.5$:

$$P(|\bar{X} - \mu| \geq 5) \leq \frac{1}{k^2} = \frac{1}{6.25} = \boxed{0.16}.$$

★ *Intuition:* Notice the variance shrinking by a factor of n when you average — this single fact, combined with

Markov/Chebyshev, is the *entire* proof mechanism behind the Weak Law of Large Numbers in Section D. You've essentially already proven WLLN; Section D just makes the $n \to \infty$ limit explicit.

Problem B4 — One-sided Chebyshev (Cantelli): a genuinely tighter bound

A strategy has mean \$0, sd \$20k. Using Cantelli's one-sided inequality $P(X - \mu \geq a) \leq \dfrac{\sigma^2}{\sigma^2 + a^2}$, bound $P(\text{loss} \geq \$30\text{k})$, and compare to the (valid, but cruder) two-sided Chebyshev bound.

Solution.

$$\text{Cantelli: } P(X \geq 30) \leq \frac{400}{400 + 900} = \frac{400}{1300} \approx \boxed{0.308}.$$

Two-sided Chebyshev ($k = 1.5$):

$$P(|X| \geq 30) \leq \frac{1}{1.5^2} \approx \boxed{0.444}.$$

Cantelli is strictly tighter, as it should be — it only has to cover one tail, not two.

★ *Intuition:* Whenever a question only cares about *one* direction (a loss, not a loss-or-gain), using plain two-sided Chebyshev leaves real precision on the table. Cantelli is the one-sided refinement purpose-built for exactly this — "risk of a large loss" questions almost always want Cantelli, not the symmetric version.

C. Jensen's Inequality

Problem C1 — Variance's nonnegativity is just Jensen

Show $\text{Var}(X) \geq 0$ follows directly from Jensen's inequality.

Solution. $g(x) = x^2$ is convex, so Jensen gives $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq \boxed{0}.$$

★ *Intuition:* A fact you've used without question since Chapter 1.4 — "variance can't be negative" — is itself just a one-line consequence of convexity. Jensen isn't a niche tool; it's quietly underneath almost every variance-related fact you already trust.

Problem C2 — Jensen and option convexity

A call option pays $g(X) = \max(X - K, 0)$, a convex function of the underlying price X. Explain what Jensen's inequality says about $\mathbb{E}[\text{payoff}]$ versus the payoff at the expected price.

Solution. Since g is convex, Jensen gives:

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]) = \max(\mathbb{E}[X] - K, 0).$$

★ *Intuition:* This is the mathematical seed of "option time value": because a call's payoff is convex, averaging the payoff *over the price's randomness* is always worth at least as much as evaluating the payoff at the price's average — and strictly more whenever the price is genuinely random. More volatility (more spread in X) only ever helps a convex payoff, never hurts it — which is precisely why options get *more* valuable as volatility rises.

Problem C3 — Jensen explains why dollar-cost averaging works

You invest \$1000/month for 3 months at prices \$20, \$25, \$16. Compare the *arithmetic mean* price to the *actual average price you paid per share*, and connect the gap to Jensen's inequality on $g(x) = 1/x$.

Solution. Arithmetic mean price: $(20 + 25 + 16)/3 \approx \boxed{\$20.33}$. Shares bought each month: $50, 40, 62.5$ — total $= 152.5$ shares for \$3000.

$$\text{Actual avg. price paid} = \frac{3000}{152.5} \approx \boxed{\$19.67}.$$

Since $g(x) = 1/x$ is convex on $x > 0$, Jensen gives $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$, i.e. the harmonic mean (what dollar-cost averaging actually computes) is always $\leq$ the arithmetic mean.

★ *Intuition:* This is a genuinely counter-intuitive, real, and profitable consequence of convexity: buying a *fixed dollar amount* every month (so you automatically buy more shares when prices dip) always beats buying a fixed *number of shares* every month, on average price paid — not because you're timing the market, but purely because $1/x$ is convex. Dollar-cost averaging is Jensen's inequality, monetized.

Problem C4 — The plug-in trap, quantified exactly

A risk model naively estimates expected quadratic loss as $g(\mathbb{E}[X])$ instead of $\mathbb{E}[g(X)]$, for $g(x) = x^2$, with $\mathbb{E}[X] = 0$, $\text{Var}(X) = 25$. Exactly how much does the naive estimate understate the truth?

Solution.

$$\text{Naive: } g(\mathbb{E}[X]) = 0^2 = 0.$$

$$\text{True: } \mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 = 25 + 0 = \boxed{25}.$$

$$\text{Understatement} = \boxed{25}$$

(exactly $\text{Var}(X)$, always, for $g(x) = x^2$).

★ *Intuition:* This is the exact same trap as Chapter 2.1's nonlinear-plug-in problem (squaring the German Tank MVUE), now revealed as a general Jensen's-inequality phenomenon rather than a one-off coincidence: plugging a mean into *any* convex function always understates the true expectation, by an amount driven by the variance. The more uncertain X is, the worse the naive plug-in gets — never the reverse.

D. Weak Law of Large Numbers

Problem D1 — WLLN from Chebyshev, explicitly

State the WLLN, and show $P(|\bar{X}_n - \mu| \geq \varepsilon) \to 0$ as $n \to \infty$ for any fixed $\varepsilon > 0$, given finite variance σ^2.

Solution. By Chebyshev applied to $\bar{X}_n$ (as in B3):

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n \to \infty} \boxed{0}.$$

★ *Intuition:* The WLLN isn't a separate, mysterious theorem to memorize — it's the $n \to \infty$ punchline of a bound you can write down from Chebyshev in one line. "More data makes the sample mean converge to the truth" is just "a shrinking fraction, σ^2/n, divided by a fixed ε^2, goes to zero."

Problem D2 — How large does n actually need to be?

A desk wants $P(|\bar{X} - \mu| \geq 0.01) \leq 0.05$, with worst-case per-trade variance $\sigma^2 = 0.25$ (a Bernoulli-type rate). Using Chebyshev, how many trades are needed?

Solution. Require $\dfrac{\sigma^2}{n\varepsilon^2} \leq \delta$:

$$n \geq \frac{\sigma^2}{\delta\varepsilon^2} = \frac{0.25}{0.05 \times 0.0001} = \boxed{50{,}000}.$$

★ *Intuition:* Fifty thousand trades just to pin down a rate

within $\pm 1\%$ with 95% confidence — Chebyshev-based WLLN guarantees are rigorous but famously data-hungry, because (like Markov) they make zero use of the actual shape of the distribution. Section E's CLT-based version of this exact same question needs *far* fewer trades (see E3).

Problem D3 — Monte Carlo sample-size planning

You're estimating an unknown probability p by simulation (worst case $\sigma^2 = p(1-p) \le 0.25$). How many independent simulations are needed so your estimate is within 0.02 of the truth with probability ≥ 0.99?

Solution.

$$n \ge \frac{\sigma^2}{\delta\varepsilon^2} = \frac{0.25}{0.01 \times 0.0004} = \boxed{62{,}500}.$$

★ *Intuition:* This is the Chebyshev-rigorous answer to a question every quant eventually asks: "how many Monte Carlo trials is *enough*?" It's deliberately the most conservative possible answer (valid for literally any underlying distribution) — in practice, people use the CLT instead and get away with dramatically fewer trials, which is exactly what Section E demonstrates next.

E. The Central Limit Theorem

Problem E1 — CLT for a sum

100 iid trades, mean \$50, sd \$200 each. Approximate $P(\text{total P\&L} > \$6000)$.

Solution. $S_{100} \approx \mathcal{N}(100 \times 50,\ 100 \times 200^2) = \mathcal{N}(5000,\ 4{,}000{,}000)$, sd $= 2000$:

$$Z = \frac{6000 - 5000}{2000} = 0.5$$

$$\implies P(S > 6000) \approx 1 - \Phi(0.5) \approx \boxed{0.309}.$$

★ *Intuition:* Note we never needed to know the *shape* of each trade's P&L distribution — CLT says the sum of 100 of *anything* with finite variance looks Normal, regardless of the ingredients. This universality is what makes CLT the workhorse of practical risk estimation.

Problem E2 — CLT for an average

Same setup as E1. Approximate $P(\text{average P\&L per trade} < \$20)$.

Solution. $\bar{X} \approx \mathcal{N}\left(50,\ \frac{200^2}{100}\right) = \mathcal{N}(50, 400)$, sd $= 20$:

$$Z = \frac{20 - 50}{20} = -1.5 \implies P(\bar{X} < 20) \approx \Phi(-1.5) \approx \boxed{0.0668}.$$

★ *Intuition:* Same underlying randomness as E1 (literally the same 100 trades), just asked as an average instead of a sum — always the identical CLT machinery, only the normalization (divide the sum's variance by n^2, or equivalently divide the sum's sd by n) changes.

Problem E3 — CLT vs. Chebyshev: the same guarantee, dramatically fewer samples

Redo D2's question (within 0.01, 95% confidence, worst-case sd 0.5) using the CLT-based Normal approximation instead of Chebyshev. How many trades are actually needed?

Solution. Need $z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} \le \varepsilon$, i.e. $1.96 \times \frac{0.5}{\sqrt{n}} \le 0.01$:

$$\sqrt{n} \ge \frac{1.96 \times 0.5}{0.01} = 98 \implies n \ge 98^2 = \boxed{9{,}604}.$$

Compare to Chebyshev's $50{,}000$ — CLT needs $\approx \boxed{5.2\times}$ fewer samples for the *identical* guarantee.

★ *Intuition:* This is the payoff for trusting the Normal approximation once n is reasonably large: the same 95% confidence now costs roughly a fifth of the data. Chebyshev's guarantee is airtight for literally any distribution; CLT's is an approximation that kicks in for large n — in practice, quants almost always use CLT sizing and reserve Chebyshev for when they genuinely can't assume n is large enough yet.

Problem E4 — CLT on a completely different distribution

Daily trade counts are iid Pois(λ=4). Approximate $P(\text{total trades over a 90-day quarter} > 380)$.

Solution. $S_{90} \approx \mathcal{N}(90 \times 4,\ 90 \times 4) = \mathcal{N}(360, 360)$ (Poisson: Var=mean), sd ≈ 18.97:

$$Z = \frac{380 - 360}{18.97} \approx 1.054$$

$$\implies P(S > 380) \approx 1 - \Phi(1.054) \approx \boxed{0.146}.$$

★ *Intuition:* Poisson counts look nothing like a bell curve individually (especially for small λ) — but summed across 90 independent days, CLT takes over anyway. This is the same universality from E1, now demonstrated on a genuinely different, distinctly non-Normal-looking building block.

Problem E5 — The continuity correction: a detail worth getting right

Redo E4's calculation for $P(S_{90} \ge 380)$ *exactly* (not > 380), using the continuity correction (treat the boundary as 379.5 for a discrete sum). How much does this change the answer?

Solution.

$$Z_{\text{cc}} = \frac{379.5 - 360}{18.97} \approx 1.028$$

$$\implies P(S \ge 380) \approx 1 - \Phi(1.028) \approx \boxed{0.152}.$$

Versus E4's 0.146 — a difference of about $\boxed{0.006}$.

★ *Intuition:* Small, but real: a discrete sum (trade counts are always whole numbers) has probability mass sitting exactly *at* 380, which "$S > 380$" silently excludes but "$S \ge 380$" should include. The continuity correction is the standard fix — shift the boundary half a unit toward the center of the distribution before applying the continuous Normal approximation. At Stage-1 speed this half-point rarely changes your final MCQ answer, but on a numeric-entry question it absolutely can.

Drill order for review: re-derive A2 (Markov→Chebyshev), C3 (Jensen→dollar-cost averaging), and E3 (Chebyshev vs. CLT sample sizing) from scratch, closed-book. Together they carry this chapter's central arc: every bound here is provably valid but at a cost — Markov needs nothing but is loosest, Chebyshev adds variance and tightens, Jensen exploits convexity structure, and the limit theorems trade worst-case rigor for the much sharper (but asymptotic) power of large n.